

Week 4, Day 2 : Preprocessing & Model Comparison

Preprocessing

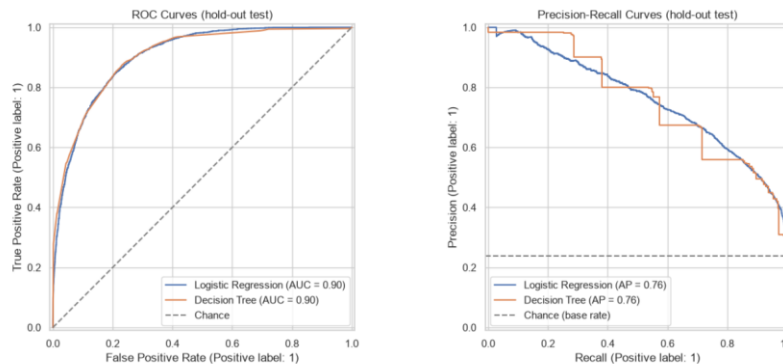
Numeric features (age, education-num, capital-gain, capital-loss, hours-per-week): median imputation -> StandardScaler. Median chosen over mean for robustness to skew/outliers in capital-gain/loss. fnlwgt (a Census sampling weight, not a person-level feature) was dropped; removing it changed all metrics by <0.001, empirically confirming it carried no signal. Categorical features: most-frequent imputation -> one-hot encoding (handle_unknown='ignore'). A log1p transform on capital-gain/loss was tested to address skew but was reverted because it measurably worsened Logistic Regression (F1 0.6576 to 0.6417, ROC AUC 0.9040 to 0.8960) while leaving the Decision Tree unchanged, since the feature's signal appears to be a threshold effect (any gain vs. none) that log-compression blurs rather than clarifies.

Model Comparison (Hold-Out Test Set)

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Majority-class	0.7607	0.0000	0.0000	0.0000	0.5000	0.2393
Rule-based (edu-num $\geq$ 13 OR cap-gain>0)	0.7452	0.4746	0.6031	0.5312	0.7426	0.4946
Logistic Regression	0.8501	0.7264	0.5997	0.6570	0.9038	0.7598
Decision Tree	0.8578	0.7938	0.5483	0.6486	0.9048	0.7597

Both models clear the rule-based baseline by a wide margin, roughly doubling F1 and reaching ROC AUC ≈ 0.90 / PR AUC ≈ 0.76 (vs. a ~ 0.24 chance floor given the $\sim 24\%$ positive rate).

ROC & Precision-Recall Curves



Error-type discussion. At the default 0.5 threshold, both models produce more false negatives than false positives. Given Day 1's framing — false positives (wasted outreach) are the costly error, so we lean toward precision and are willing to tolerate some missed high-earners — this is the right direction: both models are already conservative about who they flag as $\geq \$50K$. If recall becomes more important later, Day 3 can lower the classification threshold to trade some of that precision back for recall.

Both models trace nearly identical curves - ROC AUC and PR AUC are statistically tied. At the default 0.5 threshold both are more conservative (more false negatives than false positives), which aligns with the project's precision-first goal (false positives = wasted outreach). The Decision Tree's PR curve is stepped due to its limited leaf count (102), while Logistic Regression's is smooth from continuous probabilities.

Model Selection & Next Steps

Continuing with both models into Day 3. Logistic Regression is the interpretable, production-friendly candidate because its coefficients (capital-gain, marital status, education-num as top drivers) directly validate Day 1's manual error analysis. The Decision Tree captures nonlinear/threshold interactions a linear model cannot (e.g., splitting on capital-gain twice within one branch), while its 0.0023 train/test accuracy gap confirms it is not overfitting at depth 8. Since both are statistically tied on AUC, the deciding factor for Day 3 is not raw performance but which further tuning (regularization, depth/pruning, class weighting, threshold selection) shifts precision further in the desired direction; a Random Forest / Gradient Boosting model will also be tested as a stronger nonlinear baseline.

Planned for Day 3: binarized has_capital_gain flag (capital-gain > 0), class weighting, regularization/depth tuning, and raising the decision threshold above 0.5 to trade recall for precision.